

environments for occupants, and improved energy and water efficiency, among other benefits.

## ***Making buildings energy-efficient.***

The first step towards creating a green building is using passive temperature control methods such as solar, designing based on regional climates, passive heating and passive cooling. Simple methods like natural daylight saving designs, and windows for ventilation and insulation to control heat can help conserve energy.

Using energy-efficient appliances is another smart way of contributing to the reduction of harmful power plant emissions caused due to the burning of coal. These appliances might come with higher installation costs but a significant decrease in electricity bills will balance it off.

Green buildings should use renewable sources of energy for power, including solar, wind and thermal. Considering its availability and cost-effectiveness, solar PV technology is the most ideal source of energy for buildings. Solar panel installation can provide up to 40 per cent cheaper electricity, as claimed by Tata Power Solar. Prices for solar installations vary based on requirement. Some other providers are Sunfuel Technologies, Solar India Enterprise and Solar Universe India.

Samsung R&D Centre in Bengaluru recently adopted solar technology for its power requirements. It now draws 88 per cent power from the solar farm. Since December 2018, Samsung has drawn millions of units of solar power, instead of depending on energy derived from traditional sources.

HMR Institute of Technology & Management has installed a 146kW solar rooftop system that generates 204,000 units of electricity annually, which will offset 3679 tonnes of carbon over the next 25 years.

There are various other corporate buildings, institutions and residential societies that are reducing their carbon footprint using solar technology, reducing the burden on electrical grids.

**Green materials.** The most prevalent building material is concrete. It is highly polluting and one of the primary producers of carbon dioxide—a greenhouse gas that is the prime reason behind global warming. In view of this, the construc-

## **Green buildings in India**

Suzlon, One Earth in Pune is an excellent example of sustainable buildings in India. The company has created a green campus and employed rainwater harvesting, off-site waste conversion, renewable energy source and maximised daylight use to save energy. It claims to have reduced its operating costs by 35 per cent. It also uses several technologies to reduce brightness and cooling requirements based on the availability of natural light and temperature.

Infosys campus in Mysore has also incorporated sustainable site development, water savings, energy efficiency, materials selection and indoor environmental quality, and achieved 40 per cent energy savings.

tion industry is revolutionised with the advent of green materials, which can significantly reduce carbon emissions. Production, placement and the entire lifecycle of green materials have a much lower environmental impact as compared to concrete.

Additionally, green materials can help achieve a healthier indoor environment by improving air quality. These are durable, cost-effective and easily-available. Straw bale, rammed earth, hempcrete, bamboo, recycled plastic and wood are a few wise choices as an alternative to concrete.

There are numerous green building materials available for construction. Before deciding which one(s) to use, consider their availability and feasibility in local regions, material safety data sheets (MSDS), indoor air quality (IAQ) test data, product warranties, recycled content information, durability data and government regulations.

The Shanghai Tower in China is constructed from locally-sourced materials, some of which included recycled materials. The building is curved to optimise wind load. The double-skin façade further allows additional insulation. The building also includes a roof-top garden. This building is a prime example of durable and strong green construction.

Green materials are also becoming common in households, and several believers of sustainable living have created beautiful homes in various remote locations.

**Water efficiency.** The concept of a green building not only aims at lower environmental impact during

construction but also requires effective upkeep throughout the lifecycle of the building. This means that the building should function with minimum resources. Fresh water is limited on earth and, with increasing population and water needs, there is a crucial need for saving it. Water conservation implies responsible use of water and minimising water wastage.

There are multiple technologies that can help improve water efficiency of buildings. Rainwater harvesting is one of the most prevalent ones, as it enables storing rainwater collected from the roof and using the same for domestic purposes later. Bullitt Centre, Washington, USA, uses rainwater harvesting and collects 56,000 gallons of water daily. This water is treated and supplied to meet the water needs of the building.

Companies like Osmosis India, KRG Indian and D&D Ecotech Services can help set up a rain water harvesting system.

Grey water harvesting is another common method. It treats the water before it is released to reduce environmental impact.

Additionally, in commercial buildings, water pressure is high throughout the building. With higher pressure, water flows through the system with heavy flow through the terminal fixtures beyond rated flow capacities. This additional water is often wasted. Pressure reducing valves can help save this water, using a technique called pressure reduction.

Such conscious steps will play a vital role in improving the environment and help us pass on a greener earth to future generations. We need to take responsibility of what we have neglected for so long, while creating hazardous products and technologies without considering their impact on the environment. Unknowingly, we have created a hole in the ozone layer, which protects us from UV rays that can cause a number of deadly diseases. We have a huge pacific garbage patch where aquatic life can hardly survive, and we live in extremely-polluted metropolitans where respiratory diseases are common. It is the need of the hour to make responsible choices towards building a self-sufficient space. **EFY**

—Deeksha Sharma